The other way round: from semantic search to collaborative curation

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Abstract: Society is becoming increasingly digitized. In a joint project of expert collaborators (Hochschule Darmstadt, University of Applied Sciences, TU Darmstadt, University of Technology, University + State Library Darmstadt, Software AG, media transfer AG, nterra GmbH, House of IT) the Städel Museum in Frankfurt (Germany) developed a cloud-based exhibit platform realising a concept we call "digital strolling". Redefining digital visits to a museum, the semantic search engine intuitively matches digitised multimedia content which is based on a complex system indexing unstructured and structured data related to our exhibits. The query process goes beyond typing a simple search string and browsing through a bewildering list of results. Digital visitors experience shift from "searching" to "finding" - from a narrow-minded matching of question and answer to a self-conducted process of learning and enjoying the jump into the rich and complex world of human nature seen through art. They can save their discovery paths, share and discuss them with others. This moves towards high-quality consumer curation. Though not being comparable to art historians' curations, it helps digital visitors express themselves better than submitting Facebook "likes" or pictures on Pinterest and alike. This method encourages cultural institutions to communicate with digital visitors as equals by transcending traditional barriers and promoting wide-ranging discussions of the content.

Keywords: digital exhibit platform, cloud-based, digital strolling, semantic search, collaborative curation, multimedia content, self-conducted learning

Founded in 1816, the Städel in Frankfurt is not only amongst the most well renowned art museums in Germany but is also the oldest private museum foundation there. The collection contains around 3,000 paintings, 100,000 drawings and prints, 1600 photographs and 600 sculptures. It encompasses more than 700 years of art history (from 1300 to the present). The founder sought a holistic approach, so the foundation also runs a library with around 100,000 books, and an academy for contemporary fine arts (Städelschule).

The Städel has become nationally recognized for its expertise in serving all age groups and every sector of society. Activities in the museum include: typical offerings for children; special programs for elder people and the unemployed; after-hour lounges where business people can meet, have a drink and listen to a lecture; activities and guidance for teenagers, where personal skills and competences are developed with a view towards professional life (Sommerakademie [summer academy]); along with special workshops and lectures held by non-art historians who talk about themes and artworks from the perspective of their own professional domain (Gastkommentar [guest's notes]).

Our audience now includes a broad range of visitors, from children and school groups to adults, families, students, and senior citizens from all geographical origins and sectors of society. The highly diverse needs of today's audiences require us to provide an equally diverse spectrum of presentation and communication offerings that combine education and entertainment. Not only must museum activities draw on different interests and levels of knowledge as a starting point for individual access, but they also must draw on various occasions and motivations for visiting a museum. Together with a wonderful aesthetic experience, the collection's interdisciplinary approach to communication

establishes a relationship between the art and its historical, literary, scientific, and political context. At the same time, it serves as the basis for topical debates concerning all social aspects of life.

Nonetheless, the Städel museum faces several challenges. Art museums are symbolic locations with alternative concepts of time and reality, they are places to pause, slow down, and focus - places where we can experience a unique work of art or even an entire collection in person. However, society is becoming increasingly digitized, and it is not stopping for the sake of physical cultural content. As a result, museums, like all institutions, are confronted by enormous changes in almost every area of life that are fundamentally redefining how they handle information, education, and culture. How can we attract people following the digital lifestyle? As the world is more and more seen through digital devices, we need to be able to make our unique content visible there. How can we deliver information in a way that addresses the different expectations and cultural backgrounds of digital visitors? How can we generate long-term interest in cultural heritage and make themes of art history relevant to modern life? How can we address diversification, on both a regional and global scale? How can we fulfil our educational mission as a public institution? How can we develop an information offering where users are able to find more than they were originally looking for? On an ideological level, we believe that - as every visitor is unique -; every museum visit, be it physical or digital, should also be unique as well. We seek to radically change the way a well-established art museum works today. We need to go from pure “acting” to “interaction”, from “delivering content” to “creating content”, and from streamlined offerings to personalisation.

An essential and unique selling point for this digitalisation strategy lies in the customised, needs-driven provision of content for future users. However, when extensive archives are simply made available digitally - unfiltered and without any specific communication - the benefits are usually restricted to visitors who are already well versed in the topic. The majority of our visitors do not know where to begin and what to search within the digital collection because they lack the necessary scholarly knowledge. Consequently, we must impart information to them through various means, similar to the experience of visiting a museum in person. This multifaceted approach provides digital visitors with a variety of options to access the collection. We link the nature of the experience with the museum’s mission to educate and communicate, enabling unlimited access to the cultural content beyond the museum's walls.

Through a group of expert collaborators comprised of partners from academia (Hochschule Darmstadt, University of Applied Sciences, and the TU Darmstadt, University of Technology), business (Software AG, mtG AG, nterra GmbH), and the cultural world (the library at TU Darmstadt), the Städel Museum developed a cloud-based exhibit platform for its collection, for this exact purpose. Based on current web technologies, the application will be both usable on a desktop as well as on a mobile browser. This so called responsive design makes it effective, irrespective of screen size, resolution or device. The reasoning and search mechanism is located on the server side acting as an intermediary layer to the museum databases. Our goal is to allow potential visitors from all around the world to take a "digital stroll" through our collection, with links to multimedia content for individual artists and works of art and textual background information. This self-guided path through our collection, which is oriented toward visitors’ individualised interests and learning needs, will reconfirm the relevance of the collection, of producing and sharing art, and, ultimately, of the museum itself. ‘Digital strolling’ redefines digital visits to a museum. Physically strolling through a museum means interacting with others, getting inspired and exploring new content. The Google Museum approach is not generic, as people get information on how an exhibit looks and where it resides. Transferring this information cannot replace the experience of facing the original artwork. Therefore, the value of visiting an online museum must be established on another level. The appeal of communicating culture digitally lies in the degree of insight and understanding that users are able to develop on their own by means of interactive technologies. Digital offerings should enrich the visitor’s initial interest, deliver more than requested and ignite interest in new topics. The semantic search engine achieves this by intuitively matching digitised multimedia content which is based on a complex system indexing unstructured and structured data related to our exhibits.

Consequently, our exhibit platform offers a query process that goes beyond typing a simple search string and browsing through a bewildering list of results. Instead of restraining the search process to concrete art topics such as artists, epochs or specific artworks (which might require comprehensive previous knowledge), the user is able to follow his very own personal interests. These interests may include non-artistic domains such as science, nature, social life, and non-scientific domains such as moods or emotional qualities.

Search results become part of the navigation: they consist of pictures, texts, audio and video files presented in a visually attractive and interest stimulating grid of media tiles, scaled and positioned in accordance to their estimated relevance. The relevance is continuously computed by a knowledge-based reasoning mechanism and by evaluating user behaviour. While the user is strolling through the platform's offerings, the interface adapts itself continuously according to the user's search queries and content selection. As there is no average user, media content is presented based on explicit search results, semantic relations and personal interest. The user input is conducted in a user model to ensure that each user experiences unique visits.

Therefore, the artificial intelligence computes a relevance factor for each metric. Frequently searched keywords, favourite media and content types customize the content area continuously. On a rule-based system, the user's natural language input in terms of parts of speech, named entities and syntactical structure is analysed. For the purpose of specifying resulting topics and identifying their relationships, intelligent ontologies are used, built and expanded on. Based on these ontologies, an intelligent algorithm is being developed finding how subjects relate to each other. This is the technical basis of digital art education, which develops its impact in an associative, self-conducted and extremely personalised way.

In comparison to lexical searching, the visitor gets a nudge to new but relevant content without being aware of it before. It supports digital visitors finding a path through vast offerings of digitised exhibits. Semantic search brings up results that are intrinsically relevant. For example: if a user types "Richter", the platform does not only offer him the two artists from our collection named "Richter" (Gerhard Richter and Daniel Richter), but also a "Gerichtsszene" ("Richter" in German means "judge in a court"). This already offers a huge variety of possibilities to continue the search process, not only in the way the user originally intended, but also in other directions. Visually stimulated by unexpected search results, he may spontaneously change his mind and follow other topics he is interested in. Perhaps he had never heard of Gerhard Richter and navigates now towards several of his works in our collection. Being presented with more artworks by the same artist is a common feature of digital platforms in other Museums, but semantic searching is far superior in transferring valid principles of physical art education in the digital space. If the user continues his "Richter" search by clicking on the painting "Kahnfahrt" (Boat Trip) he will not only be offered other paintings of the same period, but also those who show a similar atmosphere like Courbet's "Wave", or they may be compared through their artistic methods, or even through a similar theme, even if the two paintings are hundreds of years apart from each other, as for example the "Embarquement to Cythera" of the rococo painter Antoine Watteau.

Digital visitors of our exhibit platform experience a shift from "searching" to "finding" things - from a narrow-minded matching of question and answer to a self conducted process of learning and enjoying the jump into the rich and complex world of human nature seen through art. Digital visitors can save their discovery paths, share and discuss them with others. This moves towards high-quality consumer curation thanks to the inner semantic logic, which is far more interesting than digital exposition, a tool which has never been proven successful in the field of digital offerings. Though not being comparable to art historians' curations, it helps digital visitors express themselves better than submitting Facebook "likes" or pictures on Pinterest and alike. This method encourages cultural institutions to communicate with digital visitors as equals by transcending traditional barriers and promoting wide-ranging discussions of the content, conducted in a much better way than when the museum acts as a proclaimer, announcing an overarching and immutable truth. In a later stage of the project, we will give curators from the museum the ability to offer their personal strolling paths as specially curated content. So, approaching the issue the other way round, we will end where others start.

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